

Can LLMs Hear Tones?

Assessing Multimodal Foundation Models for Phonetic and Tonal Transcription of Mandarin Speech

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Acronyms

IU IU International University

1 Introduction

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1.1 Motivation

Over the past few years, large language models (LLMs) have advanced at a remarkable pace. Systems that only recently struggled with basic reasoning now write code, summarize documents, and hold fluent conversations across dozens of languages '[citation needed]'. Two developments are particularly relevant for this thesis. First, leading models have become multimodal: rather than processing text alone, frontier systems such as GPT-4o, Gemini, and Claude can take audio directly as input and reason over what they hear '[citation needed]'. Second, these models have become remarkably easy to use. What once required specialized machine-learning expertise, dedicated training data, and significant engineering effort is now available through a simple prompt and a general-purpose API. In principle, anyone can now build an application around a state-of-the-art speech-processing model without any technical background in speech recognition.

This accessibility is especially attractive in the context of language learning. Mandarin Chinese is one of the most widely spoken languages in the world, and interest in learning it continues to grow for cultural, academic, and economic reasons '[citation needed]'. At the same time, learning Mandarin pronunciation is notoriously difficult for non-native speakers. Mandarin is a tonal language: the same syllable can carry entirely different meanings depending on its tone, and tones are realized not as absolute pitches but as pitch contours that may even shift in connected speech through phenomena such as tone sandhi '[citation needed]'. Beyond tones, learners struggle with phoneme distinctions that do not exist in their native language. Reliable, immediate feedback on pronunciation is therefore one of the most valuable — and one of the hardest — things a learning tool can offer.

Commercial language-learning software has tried to fill this gap, but often at a price. Many popular applications are built around paid subscriptions, lock pronunciation feedback behind premium tiers, or are funded through advertising '[citation needed]'. The feedback they provide is frequently limited to the word level and rarely diagnoses what exactly went wrong at the level of individual phonemes or tones. Notably, a growing number of these products have themselves begun to integrate LLMs into their feedback and tutoring features '[citation needed]', which raises a natural question: how good are general-purpose, off-the-shelf LLMs at the underlying task in the first place?

This is the central concern of the present work. Before an LLM can give a learner meaningful pronunciation feedback, it must first be able to hear and transcribe what was actually said - including the tones. If a multimodal model can convert spoken Mandarin into an accurate phonetic and tonal transcription, then building a pronunciation-aware vocabulary trainer becomes conceptually straight-

forward: the expected pronunciation of a target word can simply be compared against the model’s transcription of the learner’s attempt, and any deviation can be surfaced as actionable feedback. The reliability of this transcription step is therefore the foundation on which any such tool depends.

Accordingly, this thesis investigates how accurately current multimodal foundation models can transcribe spoken Mandarin into a phonetic representation that captures both segmental information (phonemes) and tones. We deliberately proceed from the easier case to the harder one: as a first step, we focus on the speech of native speakers, which provides a clearer upper bound on what these models can achieve under favorable conditions. This establishes a baseline before turning to the ultimately more challenging and more relevant case of non-native learners, whose pronunciation is exactly what a vocabulary trainer would need to evaluate. In doing so, we aim to determine whether today’s general-purpose multimodal models are already good enough — out of the box — to serve as the engine of a practical, low-cost pronunciation trainer for Mandarin.

1.2 Goal of this thesis and research questions

The overarching goal of this thesis is to assess how accurately current multimodal foundation models can transcribe spoken Mandarin into a phonetic representation that captures both phonemes and tones, and to determine whether this capability is reliable enough to serve as the foundation for a practical pronunciation-based vocabulary trainer. The guiding idea is simple: if a model can faithfully transcribe what a speaker said, then a learning tool can compare this transcription against the expected pronunciation of a target word and turn any deviation into actionable feedback. The quality of that transcription step is therefore the central object of study.

To investigate this systematically, we decompose the problem along two dimensions. The first is the input modality and speaker type, ordered from easier to harder: (i) written input only, i.e. mapping Chinese characters to their phonetic form, which isolates the model’s linguistic knowledge from any acoustic perception; (ii) native-speaker speech, which establishes an upper bound under favorable acoustic conditions; and (iii) non-native learner speech, which represents the actual target scenario of a vocabulary trainer. The second dimension is the level of granularity: the word (character) level, the phoneme level (phonetic transcription without tones), and the tonal level (phonetic transcription including tones). This yields the following research questions.

RQ1 — Grapheme-to-phonetic mapping (text only). How accurately can multimodal LLMs convert written Chinese characters into a phonetic transcription, (a) without tones and (b) with tones? This serves as a text-only baseline that measures the model’s linguistic knowledge of Mandarin pronunciation independently of audio processing.

RQ2 — Transcription of native-speaker speech. How accurately can multimodal LLMs transcribe

spoken Mandarin produced by native speakers, at (a) the word (character) level, (b) the phoneme level (phonetic transcription without tones), and (c) the tonal level (phonetic transcription including tones)?

RQ3 — Transcription of non-native learner speech. How accurately can the same models transcribe spoken Mandarin produced by non-native learners, at the same three levels (word, phoneme, tone)? This is the most relevant and most challenging condition, as learner speech is exactly what a pronunciation trainer must evaluate.

RQ4 — Comparison with established systems. How does the transcription accuracy of general-purpose multimodal LLMs compare to that of dedicated speech-recognition systems (e.g. Whisper) and established pronunciation-assessment baselines? This situates the "out-of-the-box" performance of LLMs relative to specialized tools.

RQ5 — Error structure and tonal challenges. Where do the models' errors concentrate? In particular, which tones and which phoneme contrasts are most frequently misrecognized, and do characteristic confusion patterns emerge? Understanding the structure of errors is essential for judging whether, and how, such models could be integrated into a learning tool.

Taken together, RQ1–RQ3 trace the core capability from the simplest to the most demanding setting, RQ4 places this capability in the context of existing technology, and RQ5 examines its qualitative reliability. Answering these questions allows us to assess the central practical claim motivating this work: whether today's multimodal foundation models are already accurate enough, used out of the box, to power a low-cost pronunciation trainer for Mandarin learners.

Scope. This thesis focuses primarily on the transcription capability itself, evaluated first and foremost on native-speaker speech and single-word utterances, with non-native learner speech treated as the next step toward the target application. The design and user-facing implementation of a complete vocabulary trainer, as well as longitudinal studies of learning outcomes, are beyond the scope of this work and are discussed as future work (Section [X]).

1.3 Structure of this work

The remainder of this thesis is organized as follows.

Chapter 2 (Related Work) reviews the research landscape relevant to this thesis. After describing the methodology used to identify and select the relevant literature, it traces the development of pronunciation and tone recognition from traditional, pre-Transformer approaches — such as statistical speech recognition, hidden Markov models, statistical language models, and conditional random fields — to Transformer-based methods. It then surveys the use of large language models and foundation models in language learning, with particular attention to vocabulary and pronunciation learning, and to Chi-

nese speech processing, including Chinese speech recognition and tone recognition. The chapter further discusses how pronunciation and tone recognition are evaluated and which corpora exist for assessing Chinese pronunciation. It closes by distilling the key learnings and identifying the research gaps that motivate this work.

Chapter 3 (Methodology and Experimental Setup) presents the design of the empirical study. It first gives an overview of the overall methodology and the experimental setup, then details the concrete components: the vocabulary list underlying the experiments, the setup used to evaluate tonal information, and the selection of the multimodal models under study. Finally, it defines the evaluation metrics used to quantify transcription accuracy at the word, phoneme, and tonal levels.

Chapter 4 (Results) reports the empirical findings. Following the structure of the research questions, it presents the models' performance on the text-only baseline (character-to-phonetic conversion, with and without tones) and on the transcription of spoken Mandarin, progressing from native-speaker to non-native learner speech across the word, phoneme, and tonal levels, together with the comparison against established baseline systems.

Chapter 5 (Discussion and Limitations) interprets these results, discusses what they imply for the central question of whether off-the-shelf multimodal models are accurate enough to support a pronunciation-based vocabulary trainer, and critically examines the limitations of the chosen approach, data, and evaluation.

Chapter 6 (Conclusion and Future Work) summarizes the contributions of the thesis, answers the research questions posed in Section 1.2, and outlines directions for future work — in particular the extension toward non-native learner speech at scale and the design of a complete pronunciation trainer built on top of the transcription capability studied here.

2 Related Work

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4 Results

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5 Discussion and Limitations

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6 Conclusion and Future Work

6.1 Conclusion

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6.3 Future Work

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Figure 1: Logo of IU
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References

Nussbaum, M. (1978). *Aristotle's "De Motu Animalium"*.

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